Ensemble-Based Spatial Snow Water Equivalent Estimates Using the SWE Evaluation Tool for the Rio Grande Basin

March 22, 2026

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Introduction

No spaceborne sensor can directly measure snow water equivalent (SWE). Therefore, the spatial distribution of SWE must be estimated from remotely sensed observations combined with modeling approaches. Existing gridded SWE products exhibit documented accuracy limitations across both mountainous and non-mountainous regions, resulting in a wide range of SWE estimates for any given location. To address this uncertainty, the New Mexico Office of the State Engineer (NM OSE), in collaboration with the Institute of Arctic and Alpine Research (INSTAAR), developed the SWE Evaluation Tool (SWEET). SWEET integrates multiple suitable SWE data products into an ensemble framework, with the objective of producing near-real-time spatial SWE estimates for the Rio Grande Basin that are, on average, as accurate as or more accurate than any individual product alone.

Figure 1 shows the near-real-time (March 22, 2026) ensemble SWE estimates for the Rio Grande Basin. Subsequent sections of this report present detailed ensemble-based SWE maps, along with summaries by subbasin and elevation band. In addition to the ensemble results highlighted in this report, figures and tables for each individual SWE product are also provided for download to support transparency and further analysis. These data sets and visualizations are available on the GitHub repository.

For ease of navigation, hyperlinks are provided throughout this report to help readers quickly access basin-specific sections and supporting data tables.

Rio Grande Basin SWE Overview

Rio Grande Headwaters

Upper Rio Grande

Rio Grande-Elephant Butte (Middle Rio Grande)

About SWEET

Rio Grande Basin SWE Overview

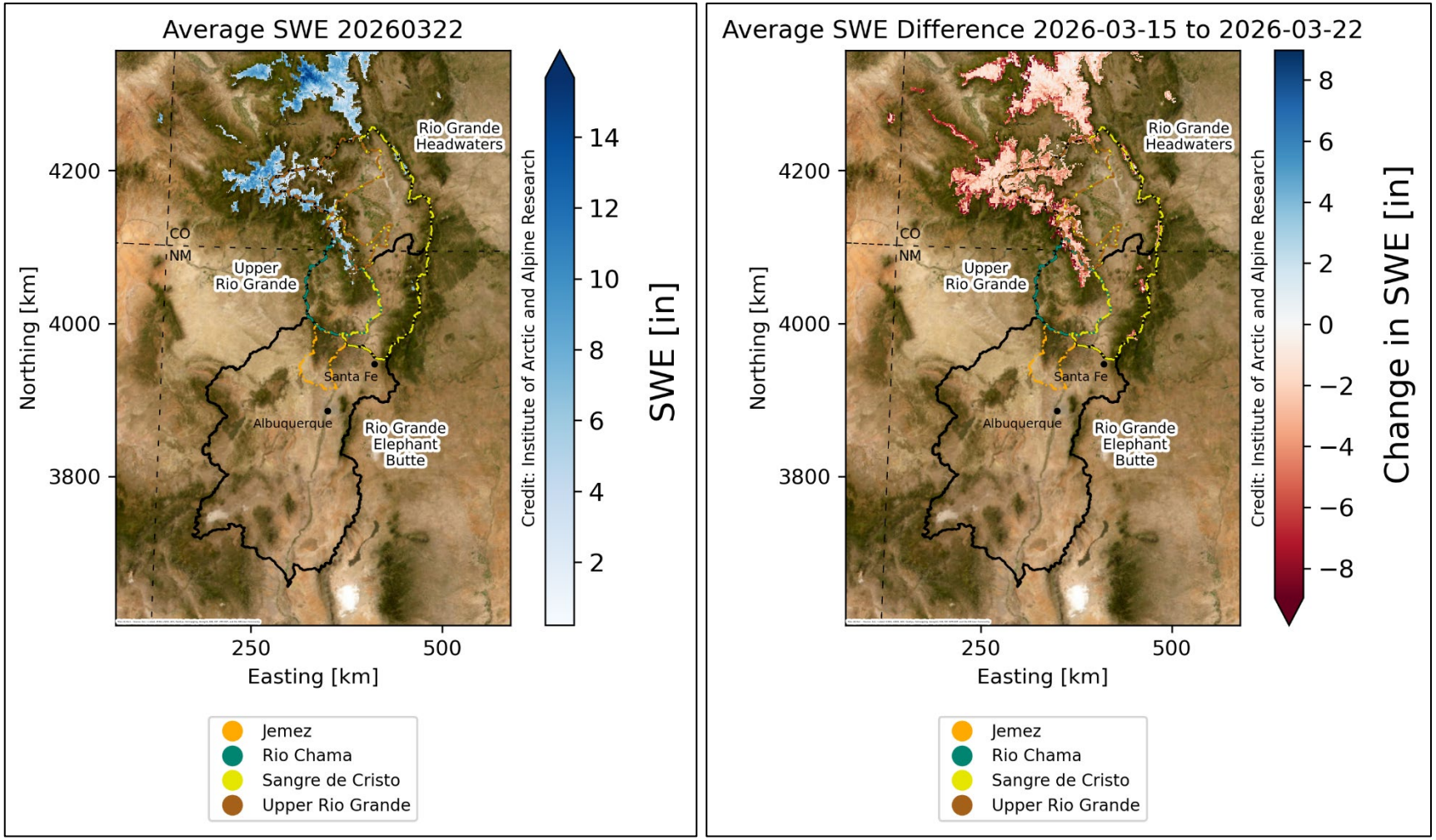

Figure 1. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Rio Grande Basin for March 22, 2026. *SWE amounts (in.) across the Rio Grande Basin (left); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Rio Grande Basin from March 15, 2026, to March 22, 2026 (right); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts.*

Table 1. Estimated ensemble SWE by HUC6 Basin for March 22, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since March 15, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value. Values are separated by HUC8 subbasin and totaled for each HUC8 basin (bold).*

HUC6 Basin	HUC8 Basin	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-03-15 (af)	Median Vol. Change since 2026-03-15 (af)
Rio Grande Headwaters	Rio Grande Headwaters	183,146	177,526	171,938 (UA_SWE)	199,974 (CU_SWE)	-118,189	-120,073
	Alamosa-Trinchera	37,057	40,982	21,303 (SNODAS)	48,887 (CU_SWE)	-41,981	-46,007
	San Luis	28,784	28,409	26,012 (CU_SWE)	31,931 (UA_SWE)	-24,616	-22,941
	Saguache	19,113	15,987	15,432 (UA_SWE)	25,919 (CU_SWE)	-32,431	-34,428
	Conejos	70,129	73,187	52,972 (UA_SWE)	84,228 (CU_SWE)	-50,013	-42,989
	Total	**338,229**	**316,413**	**313,255 (UA_SWE)**	**385,019 (CU_SWE)**	**-267,230**	**-295,331**
Upper Rio Grande	Upper Rio Grande	27,150	21,247	11,887 (CU_SWE)	48,316 (SNODAS)	-28,010	-40,387
	Rio Chama	55,882	61,972	23,891 (UA_SWE)	81,785 (SNODAS)	-53,295	-42,491
	Total	**83,033**	**73,859**	**45,138 (UA_SWE)**	**130,101 (SNODAS)**	**-81,305**	**-67,374**
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	142	39	7 (UA_SWE)	380 (SNODAS)	-412	-197
	Jemez	275	11	0 (UA_SWE)	813 (CU_SWE)	-1,269	-1,294
	Rio Grande-Albuquerque	0	0	0	0	0	0
	Arroyo Chico	0	0	0	0	0	0
	North Plains	0	0	0	0	0	0
	Rio San Jose	0	0	0	0	0	0
	Plains of San Agustin	0	0	0	0	0	0
	Rio Salado	0	0	0	0	0	0
	Jornada Del Muerto	0	0	0	0	0	0
	Elephant Butte Reservoir	0	0	0	0	0	0
	Rio Puerco	192	0	0 (SNODAS, UA_SWE)	575 (CU_SWE)	-418	-268
	Total	**609**	**392**	**7 (UA_SWE)**	**1,427 (CU_SWE)**	**-2,099**	**-2,443**

Table 2. Estimated ensemble SWE by specified subbasin groupings for March 22, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since March 15, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value. Values are separated by HUC8 subbasin and totaled for each specified subbasin grouping (bold).*

OSE Grouping	HUC8 Basin	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-03-15 (af)	Median Vol. Change since 2026-03-15 (af)
Upper Rio Grande	Rio Grande Headwaters	183,146	177,526	171,938 (UA_SWE)	199,974 (CU_SWE)	-118,189	-120,073
	Saguache	19,113	15,987	15,432 (UA_SWE)	25,919 (CU_SWE)	-32,431	-34,428
	Conejos	70,129	73,187	52,972 (UA_SWE)	84,228 (CU_SWE)	-50,013	-42,989
	Total	**272,388**	**266,700**	**240,342 (UA_SWE)**	**310,121 (CU_SWE)**	**-200,633**	**-206,704**
Sangre de Cristo	San Luis	28,784	28,409	26,012 (CU_SWE)	31,931 (UA_SWE)	-24,616	-22,941
	Alamosa-Trinchera	37,057	40,982	21,303 (SNODAS)	48,887 (CU_SWE)	-41,981	-46,007
	Upper Rio Grande	27,150	21,247	11,887 (CU_SWE)	48,316 (SNODAS)	-28,010	-40,387
	Total	**92,991**	**94,160**	**86,786 (CU_SWE)**	**98,029 (SNODAS)**	**-94,607**	**-81,284**
Rio Chama	**Total**	**55,882**	**61,972**	**23,891 (UA_SWE)**	**81,785 (SNODAS)**	**-53,295**	**-42,491**
Jemez	**Total**	**275**	**11**	**0 (UA_SWE)**	**813 (CU_SWE)**	**-1,269**	**-1,294**

Upper Rio Grande Snowpack Update Application

The OSE/ISC Upper Rio Grande Snowpack Update Application (Application) provides water managers and users with detailed SNOTEL-based SWE data organized by the same subbasin groupings shown in Table 2. The Application is updated regularly from December through May and compiles key SWE information from SNOTEL sites within each subbasin. In addition to the Upper Rio Grande groupings (Upper Rio Grande, Sangre de Cristo, Rio Chama, and Jemez), the Application also includes the San Juan River Basin, with data summarized for the Upper San Juan, Animas, and Piedra subbasins.

The Application can be viewed here[1].

[1] Please note that access to the Application may be restricted to employees of the NM OSE/ISC at this time. We are currently working on public access.

Rio Grande Headwaters

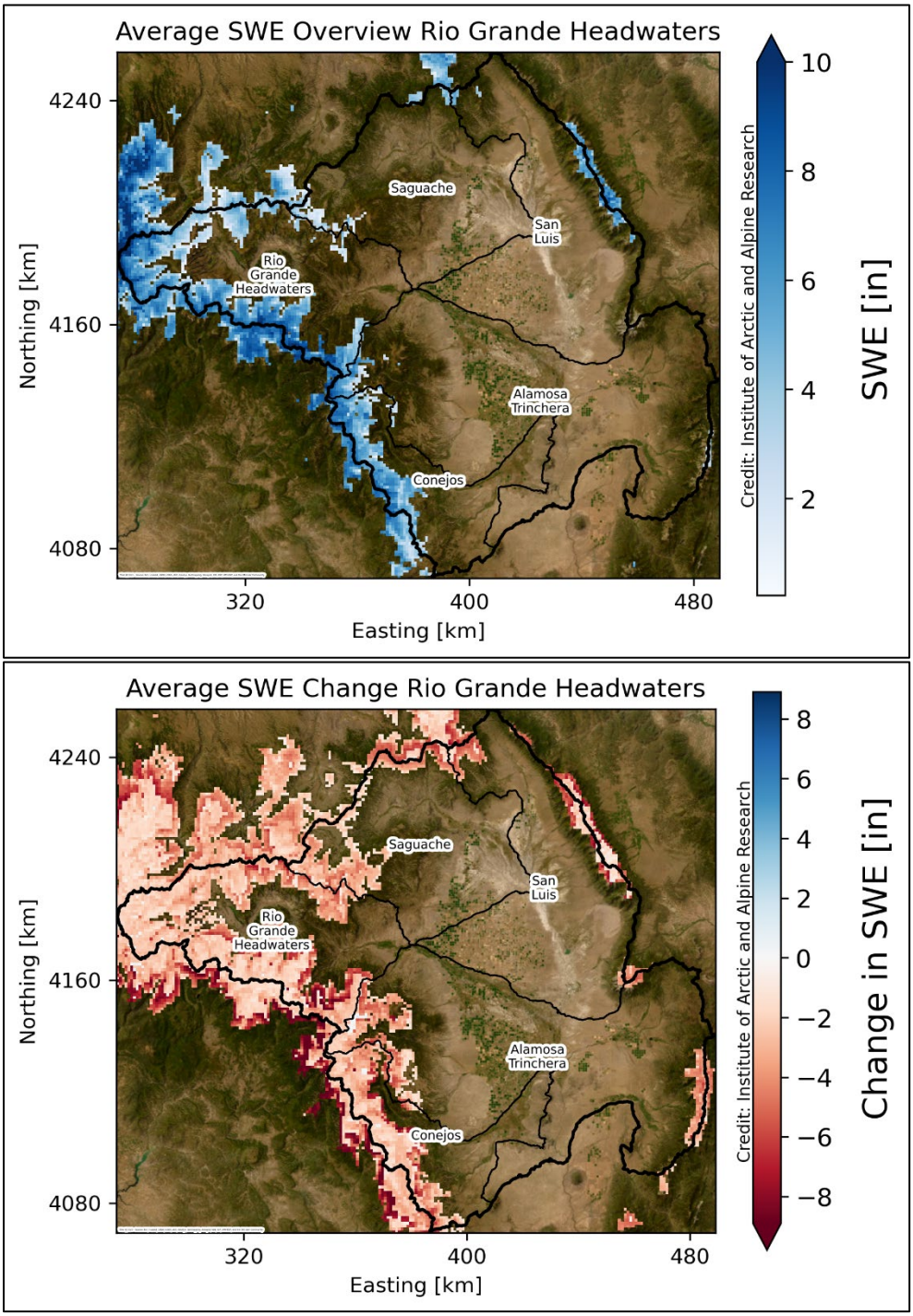

Figure 2. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Rio Grande Headwaters HUC6 Basin for March 22, 2026, derived from the SWE Evaluation Tool (SWEET). *Ensemble SWE amount across the Rio Grande Headwaters Basin, showing spatial patterns of SWE across subbasins including Rio Grande Headwaters, Alamosa-Trinchera, San Luis, Saguache, and Conejos (top); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Rio Grande Headwaters from March 15, 2026, to March 22, 2026 (bottom); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts. Basin and subbasin boundaries are shown in black.*

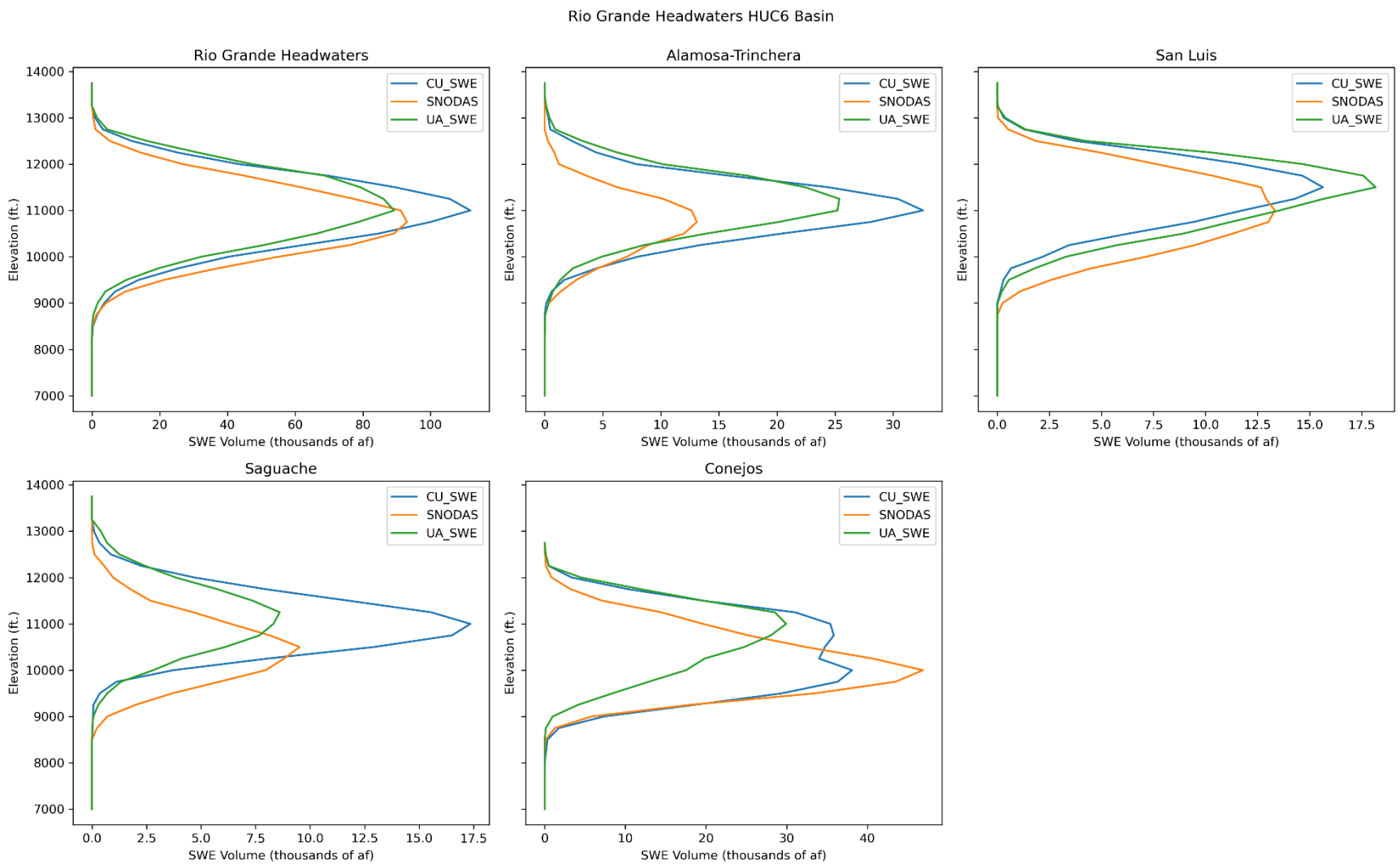

Figure 3. Estimated SWE volume (thousands of af) by subbasin and elevation band across the Rio Grande Headwaters HUC6 Basin for March 22, 2026. *Elevation-distributed SWE volume for the Rio Grande Headwaters HUC6 Basin and associated subbasins (Rio Grande Headwaters, Alamosa–Trinchera, San Luis, Saguache, and Conejos), comparing three spatial SWE products (CU_SWE, SNODAS, and UA_SWE). Curves show SWE volume summarized by elevation band, illustrating differences in both magnitude and vertical distribution among products across each subbasin.*

Table 3. Estimated ensemble SWE by HUC8 subbasin and elevation band across the Rio Grande Headwaters HUC6 Basin for March 22, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since March 15, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-03-15 (af)	Median Vol. Change since 2026-03-15 (af)
Rio Grande Headwaters	Rio Grande Headwaters	7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	-32	-3
		9000-10000'	3,113	3,600	1,653 (UA_SWE)	4,086 (SNODAS)	-9,289	-8,500
		10000-11000'	42,557	40,316	32,267 (UA_SWE)	55,088 (SNODAS)	-44,849	-41,532
		11000-12000'	97,425	91,234	89,257 (UA_SWE)	111,783 (CU_SWE)	-49,990	-50,535
		12000-13000'	39,139	43,390	26,761 (SNODAS)	47,266 (UA_SWE)	-13,754	-14,389
		13000-14000'	913	884	358 (SNODAS)	1,496 (UA_SWE)	-274	-520
	Alamosa-Trinchera	7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	280	341	142 (CU_SWE)	357 (SNODAS)	-4,120	-3,409
		10000-11000'	6,653	7,078	4,897 (UA_SWE)	7,983 (CU_SWE)	-16,833	-16,695
		11000-12000'	23,469	25,204	12,640 (SNODAS)	32,562 (CU_SWE)	-17,247	-21,155
		12000-13000'	6,406	7,880	1,225 (SNODAS)	10,113 (UA_SWE)	-3,621	-5,085
		13000-14000'	249	320	2 (SNODAS)	425 (UA_SWE)	-160	-148
	San Luis	7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	92	20	0 (CU_SWE)	257 (SNODAS)	-1,639	-1,738
		10000-11000'	4,207	3,290	2,184 (CU_SWE)	7,146 (SNODAS)	-8,616	-11,917
		11000-12000'	12,886	13,327	11,777 (CU_SWE)	13,554 (UA_SWE)	-9,207	-8,920
		12000-13000'	11,353	11,737	7,643 (SNODAS)	14,680 (UA_SWE)	-5,027	-6,349
		13000-14000'	246	313	36 (SNODAS)	388 (UA_SWE)	-127	-134
	Saguache	7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	-169	0
		9000-10000'	266	60	40 (CU_SWE)	697 (SNODAS)	-3,042	-1,650
		10000-11000'	4,834	3,731	2,812 (UA_SWE)	7,959 (SNODAS)	-15,828	-19,635
		11000-12000'	10,677	8,319	6,362 (SNODAS)	17,349 (CU_SWE)	-11,100	-11,655
		12000-13000'	3,172	3,839	970 (SNODAS)	4,707 (CU_SWE)	-2,244	-1,918
		13000-14000'	165	92	0 (SNODAS)	402 (UA_SWE)	-47	-60
	Conejos	7000-8000'	0	0	0	0	0	0

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-03-15 (af)	Median Vol. Change since 2026-03-15 (af)
Rio Grande Headwaters	Conejos	8000-9000'	5	0	0 (SNODAS, UA_SWE)	14 (CU_SWE)	-73	-66
		9000-10000'	4,724	5,829	990 (UA_SWE)	7,351 (CU_SWE)	-13,015	-12,389
		10000-11000'	34,167	38,093	17,524 (UA_SWE)	46,884 (SNODAS)	-24,884	-12,349
		11000-12000'	28,315	29,927	19,613 (SNODAS)	35,403 (CU_SWE)	-11,009	-11,004
		12000-13000'	2,919	3,366	861 (SNODAS)	4,530 (UA_SWE)	-1,032	-894

Upper Rio Grande

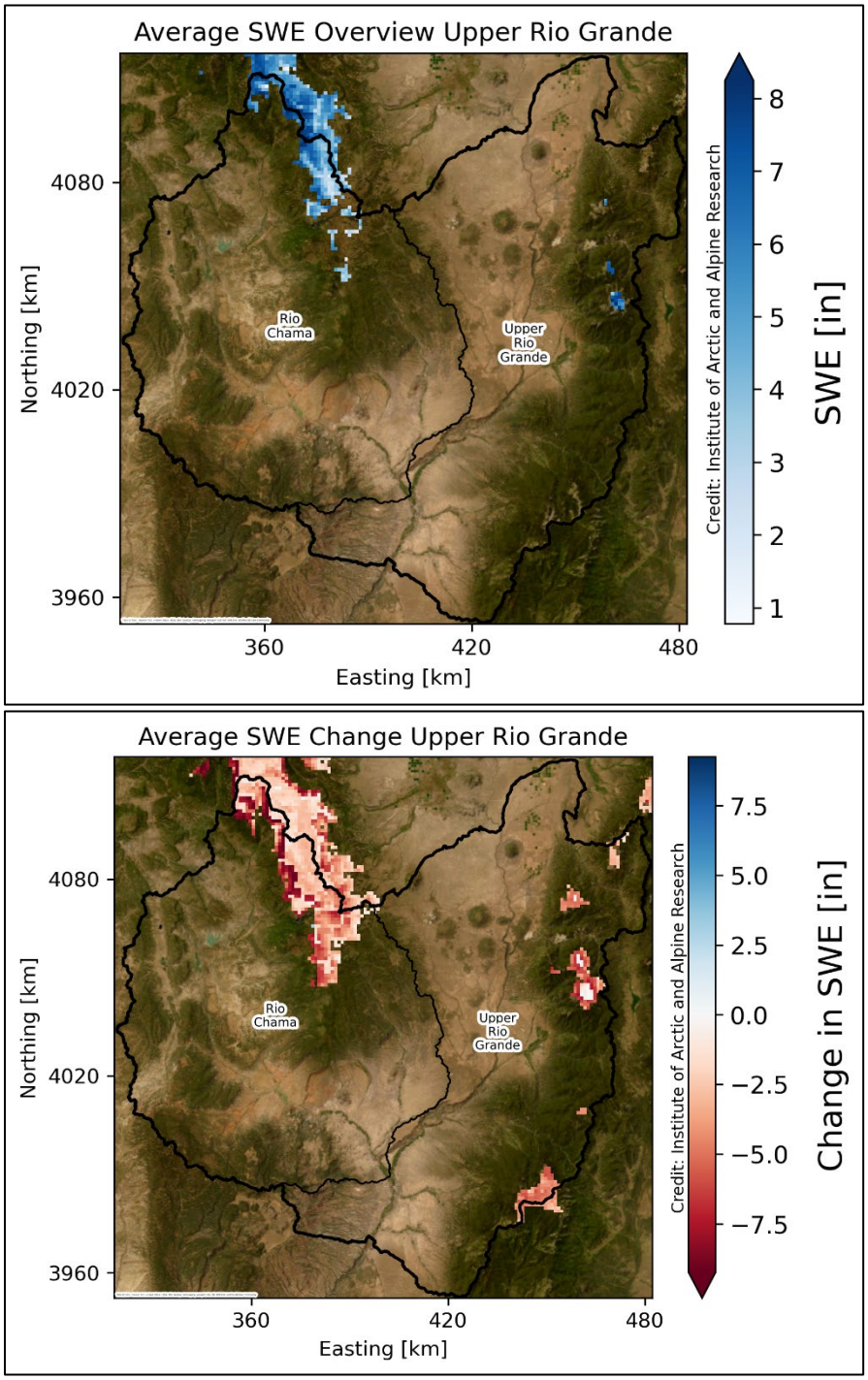

Figure 4. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Upper Rio Grande HUC6 Basin for March 22, 2026, derived from the SWE Evaluation Tool (SWEET). *Ensemble SWE amount across the Upper Rio Grande Basin, showing spatial patterns of SWE across subbasins including Rio Chama and Upper Rio Grande (top); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Upper Rio Grande Basin from March 15, 2026, to March 22, 2026 (bottom); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts. Basin and subbasin boundaries are shown in black.*

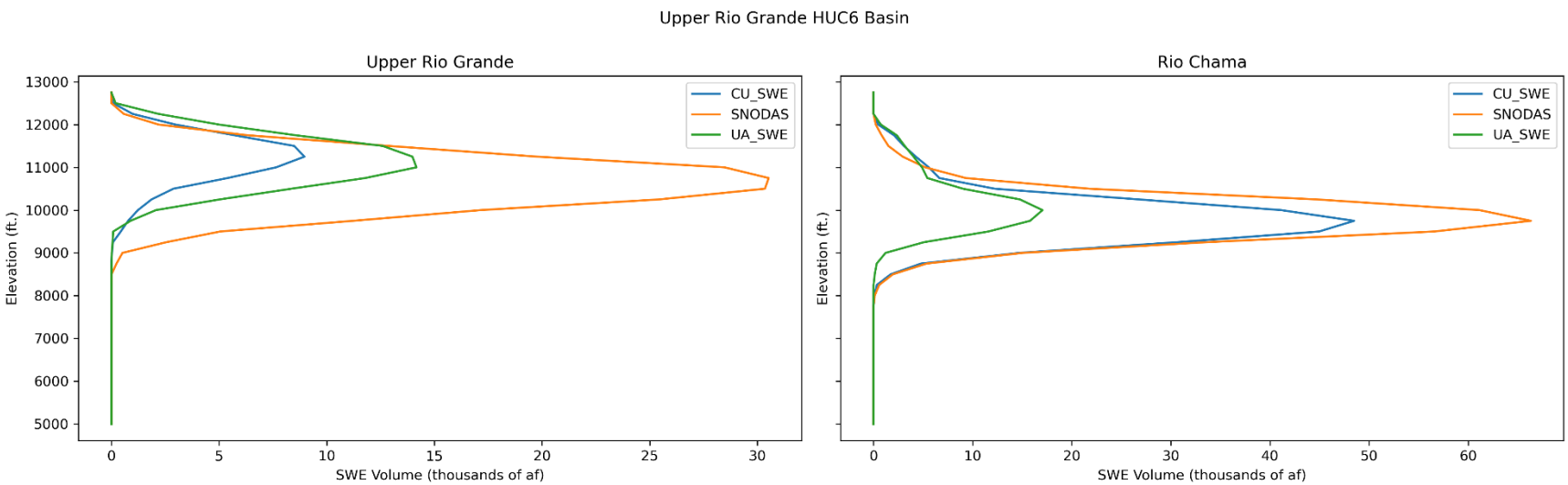

Figure 5. Estimated SWE volume (thousands of af) by subbasin and elevation band across the Upper Rio Grande HUC6 Basin for March 22, 2026. *Elevation-distributed SWE volume for the Upper Rio Grande HUC6 Basin and associated subbasins (Rio Chama and Upper Rio Grande), comparing three spatial SWE products (CU_SWE, SNODAS, and UA_SWE). Curves show SWE volume summarized by elevation band, illustrating differences in both magnitude and vertical distribution among products across each subbasin.*

Table 4. Estimated ensemble SWE by HUC8 subbasin and elevation band across the Upper Rio Grande HUC6 Basin for March 22, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since March 15, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-03-15 (af)	Median Vol. Change since 2026-03-15 (af)
Upper Rio Grande	Upper Rio Grande	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	-12	-15
		9000-10000'	187	28	18 (UA_SWE)	515 (SNODAS)	-1,479	-1,179
		10000-11000'	6,813	2,067	1,233 (CU_SWE)	17,141 (SNODAS)	-11,495	-14,615
		11000-12000'	16,764	14,168	7,652 (CU_SWE)	28,473 (SNODAS)	-12,661	-16,866
		12000-13000'	3,386	2,975	2,188 (SNODAS)	4,995 (UA_SWE)	-2,362	-4,203
	Rio Chama	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	78	111	0 (UA_SWE)	122 (SNODAS)	-1,006	-745
		9000-10000'	10,319	14,674	1,213 (UA_SWE)	15,069 (SNODAS)	-21,900	-19,396
		10000-11000'	39,778	41,140	17,049 (UA_SWE)	61,146 (SNODAS)	-28,554	-19,934
		11000-12000'	5,229	5,221	4,891 (UA_SWE)	5,576 (CU_SWE)	-1,716	-1,644
		12000-13000'	478	470	227 (SNODAS)	738 (UA_SWE)	-119	-92

Rio Grande-Elephant Butte (Middle Rio Grande)

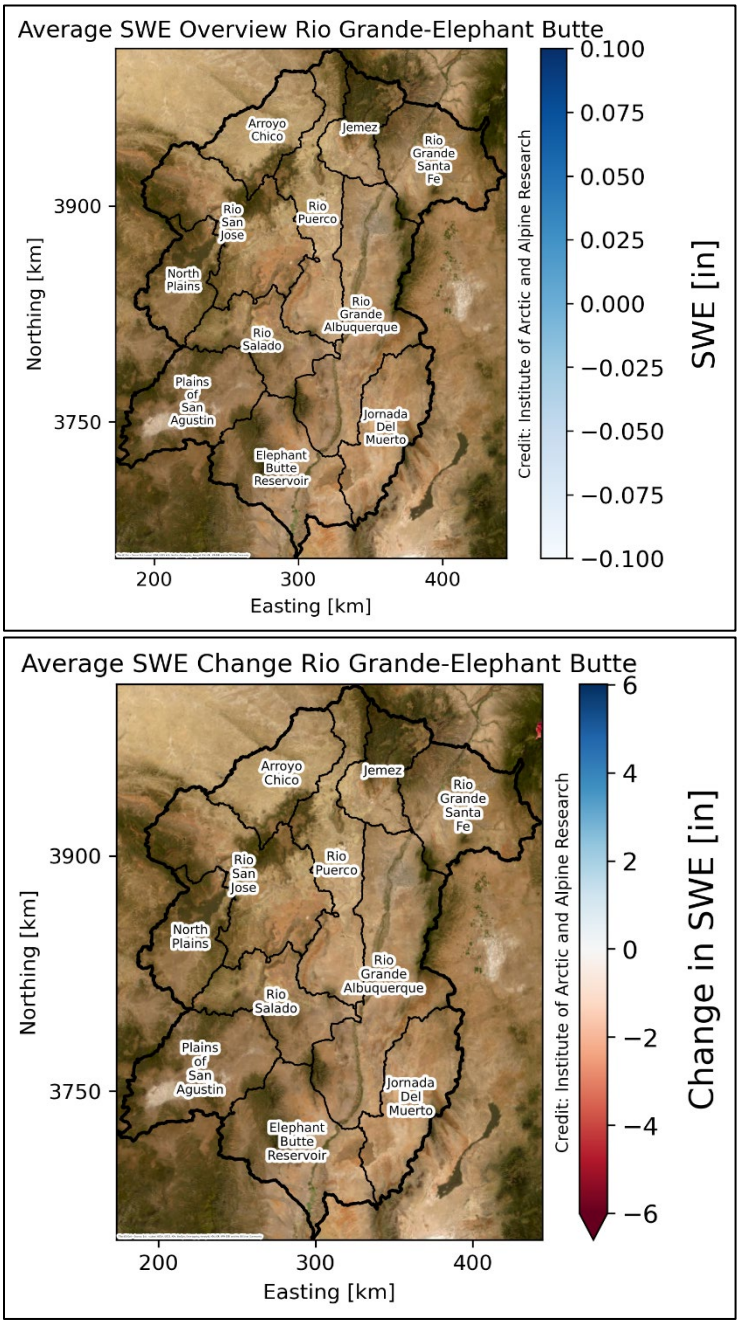

Figure 6. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Rio Grande-Elephant Butte HUC6 Basin for March 22, 2026, derived from the SWE Evaluation Tool (SWEET). *Ensemble SWE amount across the Rio Grande-Elephant Butte Basin, showing spatial patterns of SWE across subbasins including Rio Grande-Santa Fe, Jemez, Arroyo Chico, Rio San Jose, Rio Puerco, Rio Grande-Albuquerque, Rio Salado, North Plains, Plains of San Agustin, Elephant Butte Reservoir, and Jornada Del Muerto (top); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Rio Grande-Elephant Butte Basin from March 15, 2026, to March 22, 2026 (bottom); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts. Basin and subbasin boundaries are shown in black.*

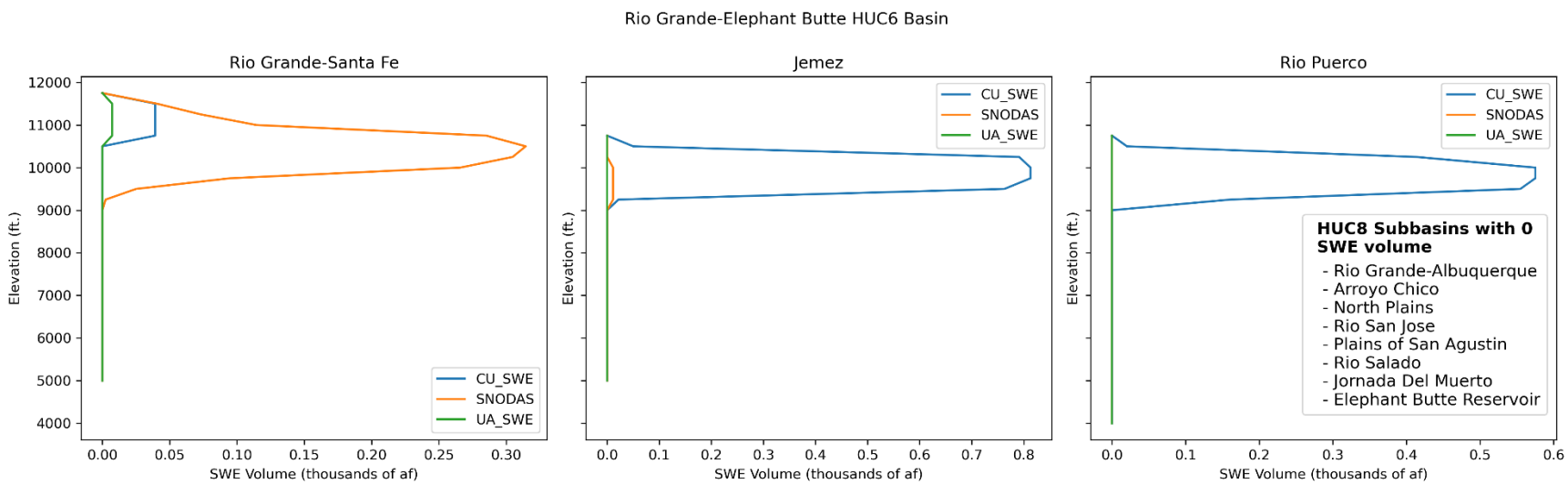

Figure 7. Estimated SWE volume (thousands of af) by subbasin and elevation band across the Rio Grande-Elephant Butte HUC6 Basin for March 22, 2026. *Elevation-distributed SWE volume for the Rio Grande-Elephant Butte HUC6 Basin and associated subbasins (Rio Grande-Santa Fe, Jemez, Arroyo Chico, Rio San Jose, Rio Puerco, Rio Grande-Albuquerque, Rio Salado, North Plains, Plains of San Agustin, Elephant Butte Reservoir, and Jornada Del Muerto), comparing three spatial SWE products (CU_SWE, SNODAS, and UA_SWE). Curves show SWE volume summarized by elevation band, illustrating differences in both magnitude and vertical distribution among products across each subbasin.*

Table 5. Estimated ensemble SWE by HUC8 subbasin and elevation band across the Rio Grande-Elephant Butte HUC6 Basin for March 22, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since March 15, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-03-15 (af)	Median Vol. Change since 2026-03-15 (af)
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	0	0	0	0	-14	0
		10000-11000'	89	0	0 (UA_SWE, CU_SWE)	266 (SNODAS)	-221	-40
		11000-12000'	54	39	7 (UA_SWE)	114 (SNODAS)	-177	-157
	Jemez	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	0	0	0	0	-453	-281
		10000-11000'	275	11	0 (UA_SWE)	813 (CU_SWE)	-816	-350
	Rio Grande-Albuquerque	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	0	0	0	0	0	0
		10000-11000'	0	0	0	0	0	0
	Arroyo Chico	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	0	0	0	0	0	0
		10000-11000'	0	0	0	0	0	0
	North Plains	6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
	Rio San Jose	5000-6000'	0	0	0	0	0	0

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-03-15 (af)	Median Vol. Change since 2026-03-15 (af)
Rio Grande-Elephant Butte	Rio San Jose	6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	0	0	0	0	0	0
		10000-11000'	0	0	0	0	0	0
	Plains of San Agustin	6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	0	0	0	0	0	0
	Rio Salado	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	0	0	0	0	0	0
	Jornada Del Muerto	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
	Elephant Butte Reservoir	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	0	0	0	0	0	0
		10000-11000'	0	0	0	0	0	0
	Rio Puerco	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
		9000-10000'	0	0	0	0	-64	-95

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-03-15 (af)	Median Vol. Change since 2026-03-15 (af)
Rio Grande-Elephant Butte	Rio Puerco	10000-11000'	192	0	0 (SNODAS, UA_SWE)	575 (CU_SWE)	-354	-170

About SWEET

The following information provides a brief overview of SWEET, and the SWE products currently included in the ensemble framework. For more detailed descriptions, citations, and additional information, please refer to the GitHub repository.

SWE products currently used include:

1. **SNODAS SWE**
    a. The SNOw Data Assimilation System (SNODAS) is a physically based snow model developed by NOAA's National Operational Hydrologic Remote Sensing Center and operational since 2004. SNODAS produces daily 1-km gridded estimates of SWE and related snow variables by combining modeled weather inputs with satellite, airborne, and in situ snow observations.
2. **UA/SWANN SWE**
    a. The University of Arizona/Snow Water Artificial Neural Network Modeling System (UA/SWANN) is a research SWE product that uses snow models, artificial neural networks, and assimilated in situ measurements to generate daily gridded SWE and snow depth. The dataset provides 4-km and downscaled 800-m SWE across the conterminous United States, incorporating SNOTEL and COOP observations with PRISM precipitation and temperature fields.
3. **CU SWE**
    a. The University of Colorado (CU) SWE-Fusion product is an operational dataset produced by INSTAAR that provides near–real-time SWE estimates at 500-m resolution. The approach fuses SNOTEL observations with historical SWE reanalyses and terrain variables using a generalized linear model, and scales results using satellite-derived fractional snow-covered area from Snow Today.

The following notes describe key processing methods, assumptions, and visualization conventions used in generating SWE Evaluation Tool figures, tables, and elevation plots.

Figures:

- For ensemble figures, CU SWE and UA SWE are reprojected to the SNODAS resolution (1 sq km) and averages are only taken at pixels where at least 2 of the 3 products have SWE values greater than 0 inches.
- All overview and subbasin plots set pixels with values of 0 inches to NA so that the basemap is visible.
- For individual product figures, each product is displayed at its native resolution.
- Elevation bands of 250 feet are used in the elevation figures, and SWE volume is calculated for each product's native resolution in each band.
- Difference plots display pixels where snow was present in either the current or previous report.
- Figures are available in PNG format on the GitHub repository.

Tables:

- For all tables, mean, median, and volume are calculated at each product's native resolution - zeros are not removed from mean and median calculations.
- For ensemble tables, mean and median volume are calculated using the total volumes from the 3 products - not from an average aggregate raster.
- For min. and max., if 2 products have 0 volume, both products are listed in that cell. If all 3 products have 0 volume, no product is listed for min. or max. in that row.
- Change in SWE will only be calculated starting with the second report of the water year.
- Ensemble calculations: For HUC8 subbasins, the ensemble (mean and median) SWE and volume change (mean and median) are calculated individually for each subbasin. For the HUC6 "Total" row (tables 1 and 2), the ensemble is recalculated using the full HUC6 basin extent rather than summing HUC8. While the ensemble mean is additive and therefore consistent between HUC8 sums and the HUC6 total, the median reflects the middle value among the three products for the entire basin. As a result, median values will not sum to the HUC6 total, since they represent the difference between basin-wide medians rather than the sum of subbasin medians.
- Tables are available in CSV and PDF format on the GitHub repository.
- Elevation tables presented in the report (Tables 3, 4, and 5) are combined into one table (Table 3) on the GitHub repository.

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